在Raspberry Pi 3上建構簡易Linux系統
Part2:Root filesystem & Test

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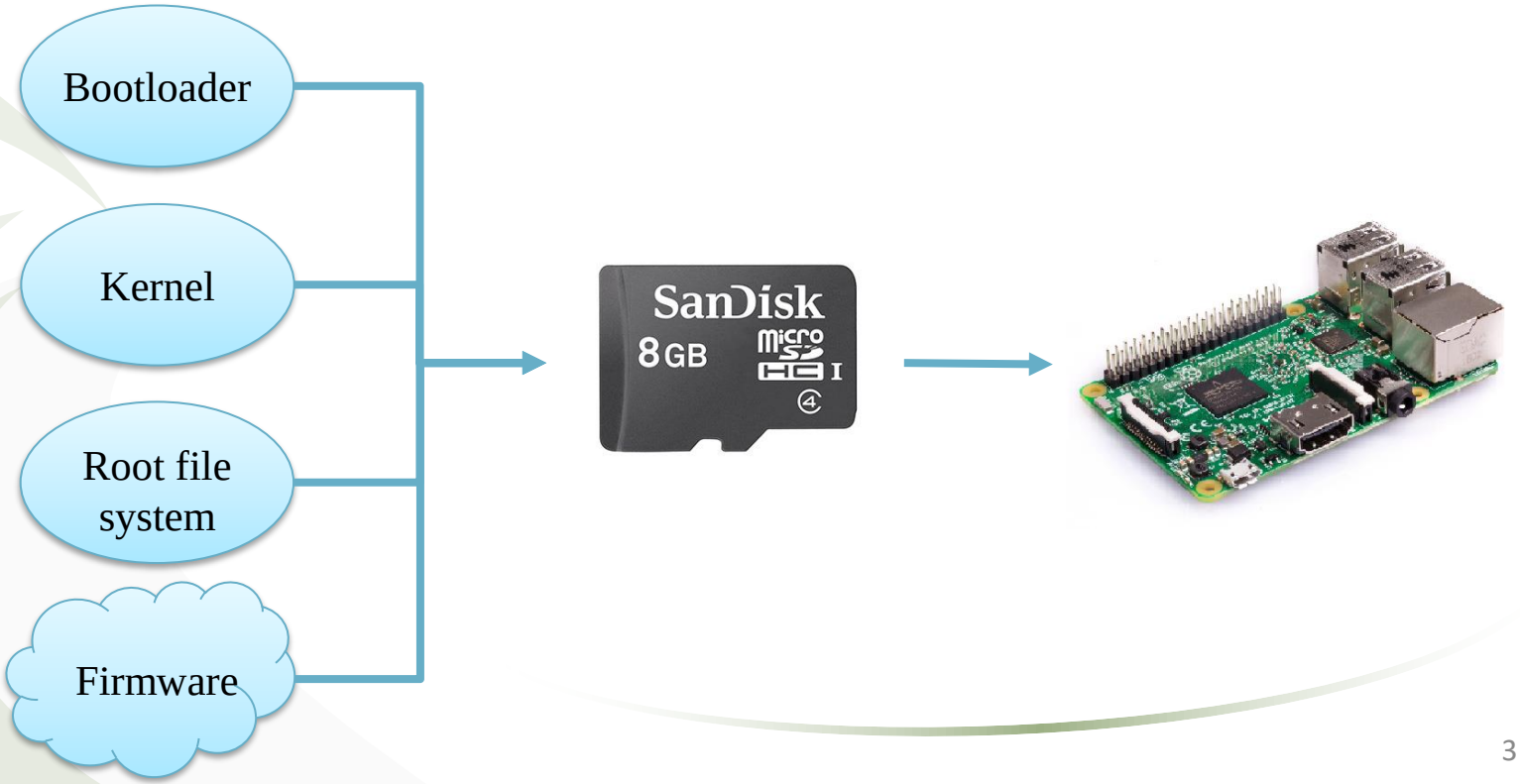
國立中正大學 資訊工程學系

Outline

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- 實驗流程
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實驗目的

- 實驗目的：
  - 於Raspberry Pi3上安裝Linux作業系統。編譯Bootloader、Kernel Image與Root file system，將其存入儲存裝置中，將Raspberry Pi 3開機。



軟硬體與環境需求

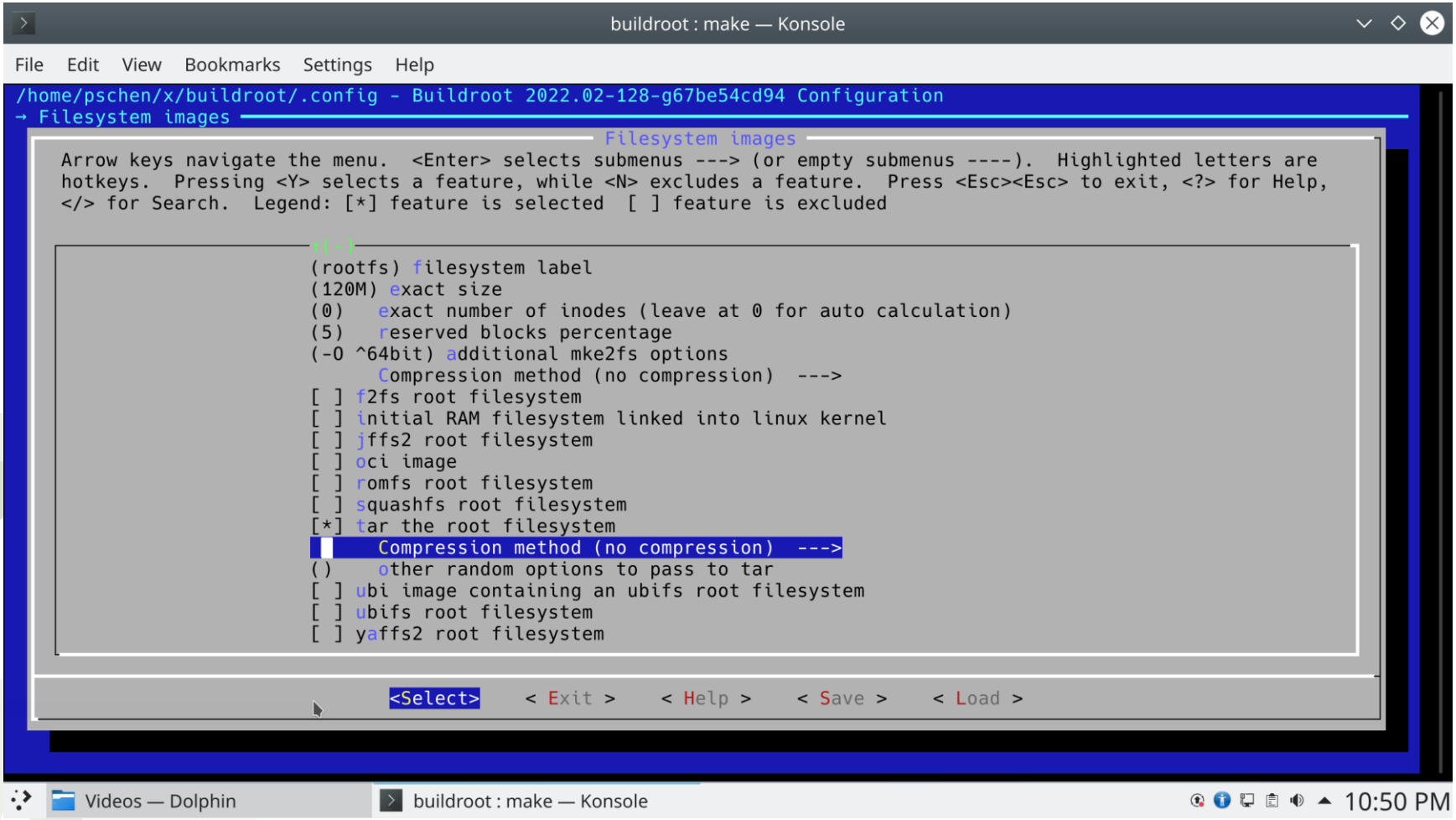
- 環境
  - Ubuntu LTS (12.04以上版本)
- Software
  - buildroot
- 硬體
  - Raspberry Pi 3 model B
  - 8G SDcard

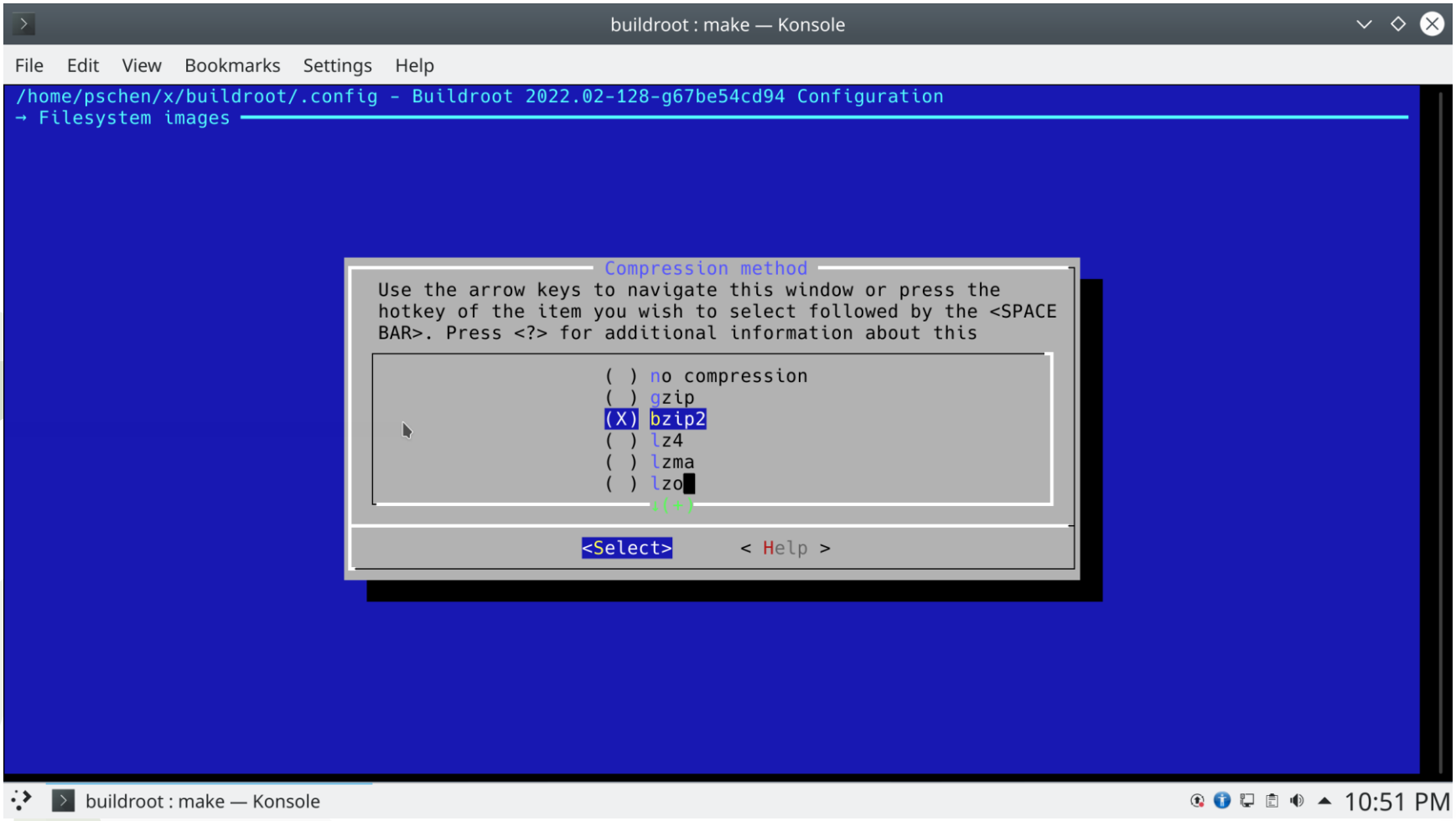
實驗流程

- 使用Buildroot編譯出：
  - Bootloader
  - Kernel
  - Root file system
- 切割並格式化SD card，將Bootloader、Kernel Image與Root file system 放入儲存裝置中
- 啟動與測試

Root filesystem

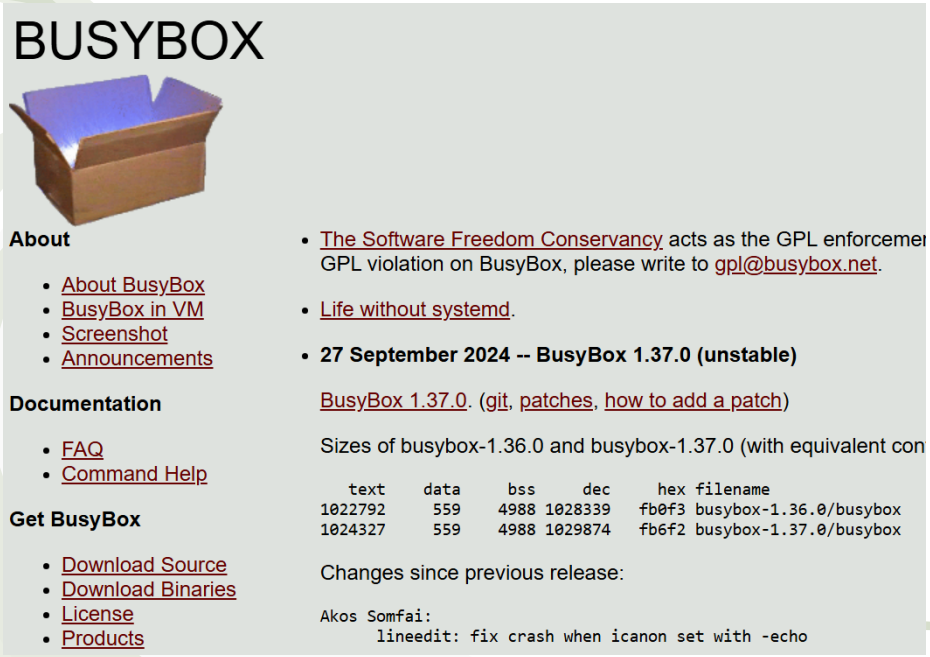
- 由於Buildroot並不是使用root權限的程式，無法直接改動權限與所有權，透過"tar the root filesystem"選項，檔案會經過一段後處理，使其能直接使用。
- **`$ make menuconfig`**
  - Filesystem images → tar the root file system → Compression method → bzip2

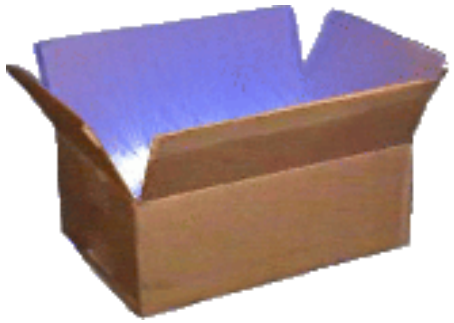

buildroot : make — Konsole
File Edit View Bookmarks Settings Help
/home/pschen/x/buildroot/.config - Buildroot 2022.02-128-g67be54cd94 Configuration
→ Filesystem images
Filesystem images
Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----). Highlighted letters are hotkeys. Pressing <Y> selects a feature, while <N> excludes a feature. Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] feature is selected [ ] feature is excluded
↑(-)
(rootfs) filesystem label
(120M) exact size
(0) exact number of inodes (leave at 0 for auto calculation)
(5) reserved blocks percentage
(-O ^64bit) additional mke2fs options
Compression method (no compression) --->
[ ] f2fs root filesystem
[ ] initial RAM filesystem linked into linux kernel
[ ] jffs2 root filesystem
[ ] oci image
[ ] romfs root filesystem
[ ] squashfs root filesystem
[*] tar the root filesystem
Compression method (no compression) --->
() other random options to pass to tar
[ ] ubi image containing an ubifs root filesystem
[ ] ubifs root filesystem
[ ] yaffs2 root filesystem
<Select> < Exit > < Help > < Save > < Load >
Videos — Dolphin
buildroot : make — Konsole
10:50 PM


buildroot : make — Konsole
File Edit View Bookmarks Settings Help
/home/pschen/x/buildroot/.config - Buildroot 2022.02-128-g67be54cd94 Configuration
→ Filesystem images
Compression method
Use the arrow keys to navigate this window or press the
hotkey of the item you wish to select followed by the <SPACE
BAR>. Press <?> for additional information about this
( ) no compression
( ) gzip
(X) bzip2
( ) lz4
( ) lzma
( ) lzo
<Select>
< Help >
buildroot : make — Konsole
10:51 PM

BusyBox The Swiss Army Knife of Embedded Linux

- BusyBox是一個精簡的Unix工具集，它統合了常用的Unix公用程式，並且高度模組化，每個功能（程式）都可以自行選擇要不要加入，又由於這些程式本來就是小型化的版本，所以BusyBox相當符合嵌入式系統對大小的需求。



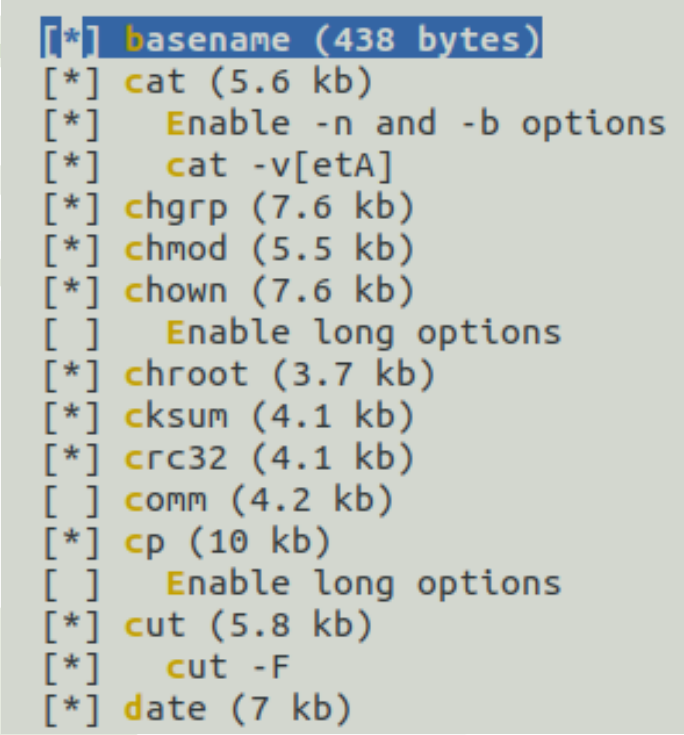


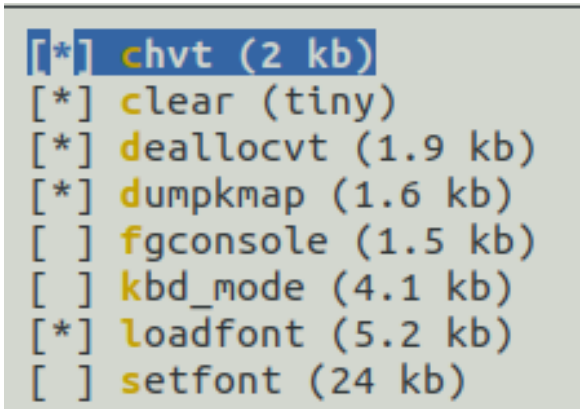
BusyBox

- BusyBox 支持超過 300 個命令，以下是部分常見類別和範例（實際支援依建置配置而定）：
  - **檔案與目錄管理**：ls、cp、mv、rm、mkdir、chmod、chown
  - **文字處理**：cat、grep、sed、awk、cut、sort、uniq
  - **系統資訊**：ps、top、free、df、uname
  - **網路工具**：ping、ifconfig、udhcpc（DHCP 客戶端）、wget、telnet
  - **壓縮與封存**：gzip、tar、bunzip2、unzip
  - **編輯與檢視**：vi、ed、less、head、tail
  - **磁碟與檔案系統**：mount、umount、fdisk、mkfs
  - **其他**：kill、mdev（裝置管理）、uevent（核心熱插拔）等
- 這些命令通常實作標準 UNIX 功能的子集，選項較簡化但足夠實用。

BusyBox

- **$ make busybox-menuconfig**
  - → Coreutils 就包含一些常用的 cat/chmod/cp
  - → Console Utilities 包含 clear/setfont





生成所有執行檔

- **Buildroot 編譯完成後，將生成包含bootloader、Linux kernel image、root filesystem。**

- **$ make -j4**
  (j後面的數字是要讓它以幾核心並行下去執行)
- Bootloader
  output/images/bcm2710-rpi-3-b.dtb
- Kernel
  output/images/Image
- Root file system
  output/images/rootfs.tar.bz2

設定哪一種壓縮方式

Firmware

- **Output/images/rpi-firmware/**
  - bootcode.bin
  - cmdline.txt
  - cofing.txt
  - fixup.dat
  - start.elf

Raspberry Pi Partitions

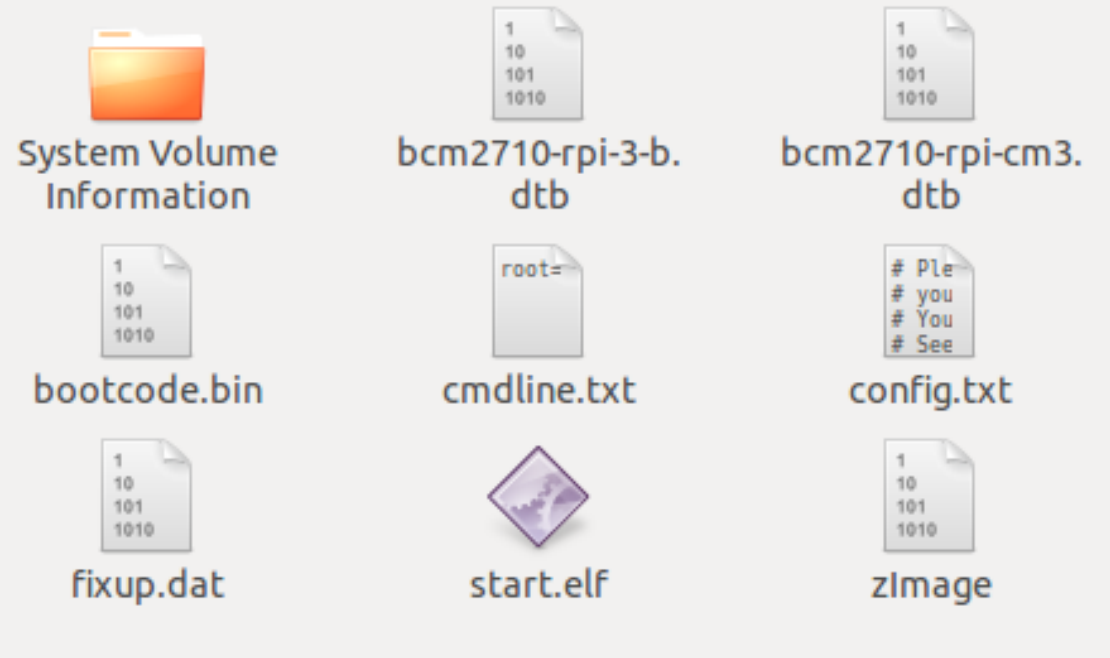
BOOT	RASPBERRY (rootfs)		
FAT32	EXT4		
• bootcode.bin • start.elf • Image • fixup.dat • config.txt • cmdline.txt • *.dtb	• /bin • /boot • /dev • /etc • /home • /lib	• /media • /mnt • /opt • /proc • /root • /run	• /sbin • /srv • /sys • /tmp • /usr • /var

Reference from https://argus-sec.com/raspberry-pi-remote-flashing/

SD card (1/2)

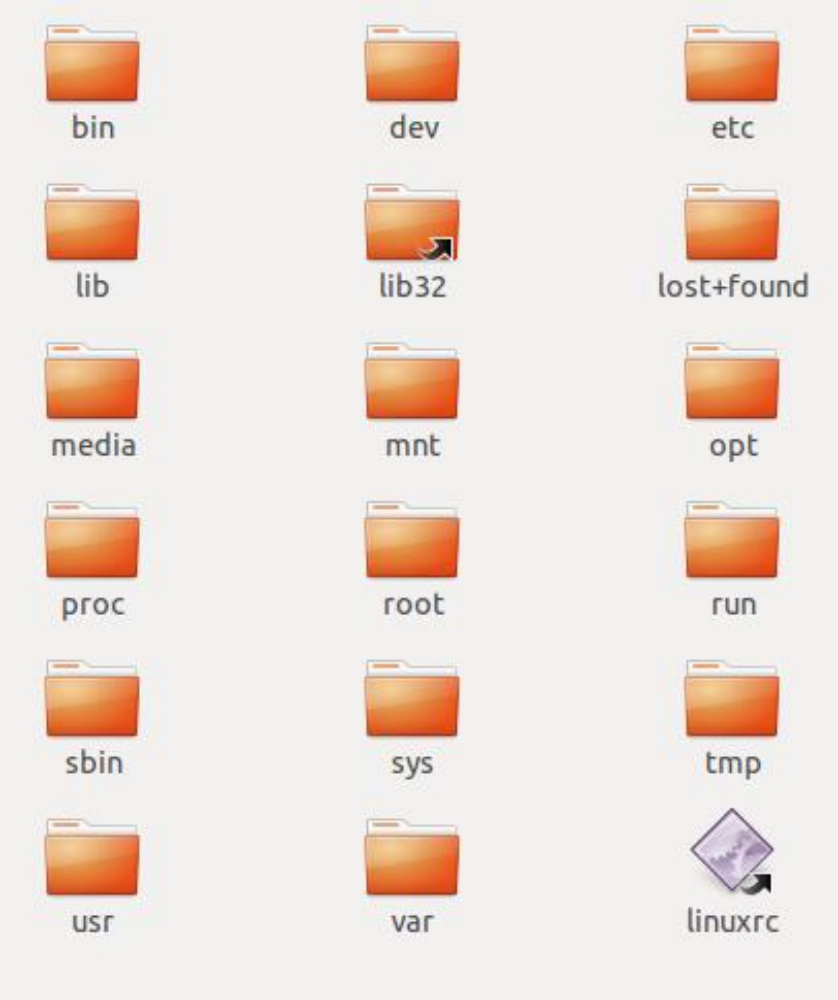
SD card的 device name

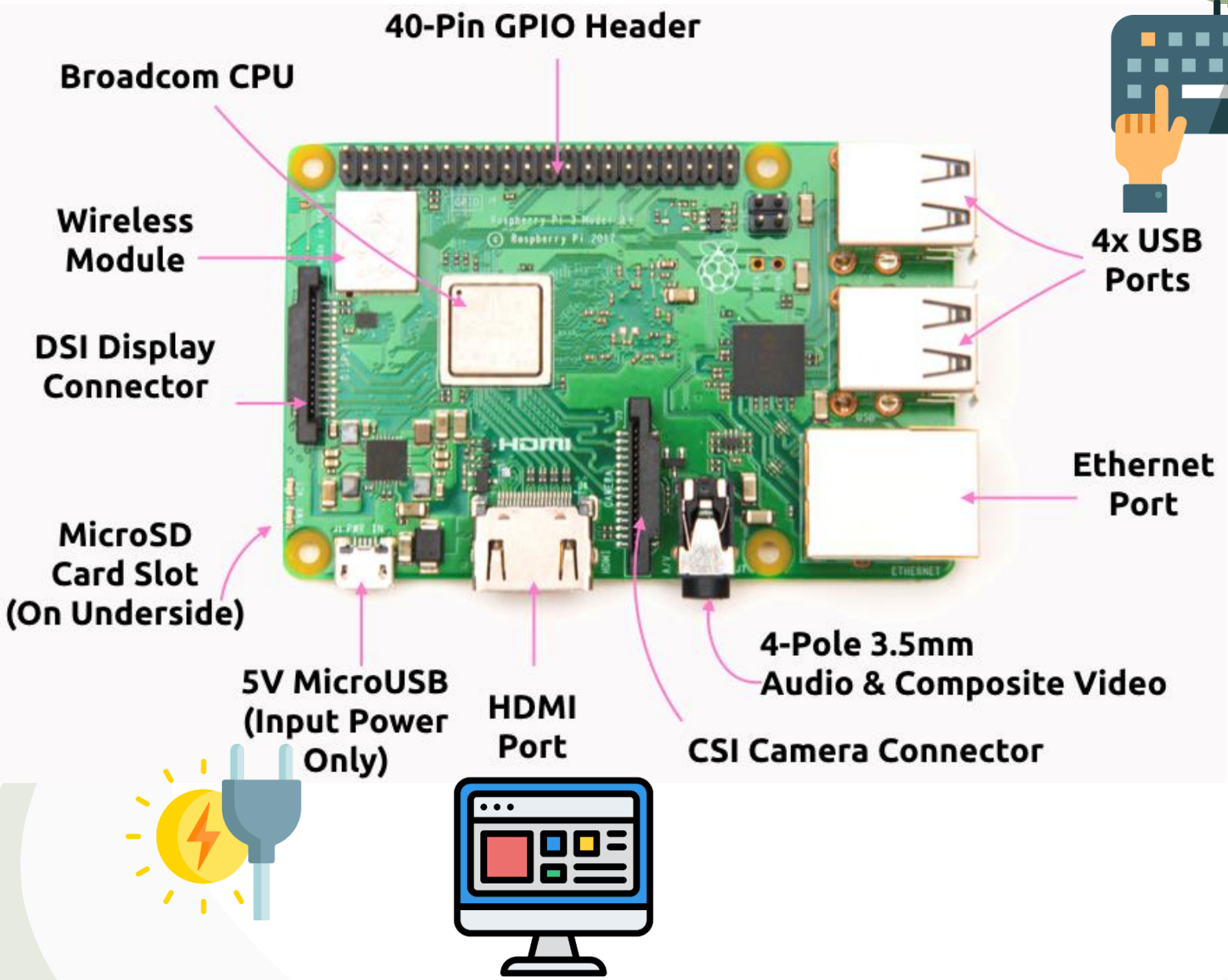
- 掛載buildroot產生出的sdcard.img來切割SD card
  - **sudo dd if=sdcard.img of=/dev/sdd**
  - 將bootloader、kernel image、與firmware寫入BOOT分割區



SD card (2/2)

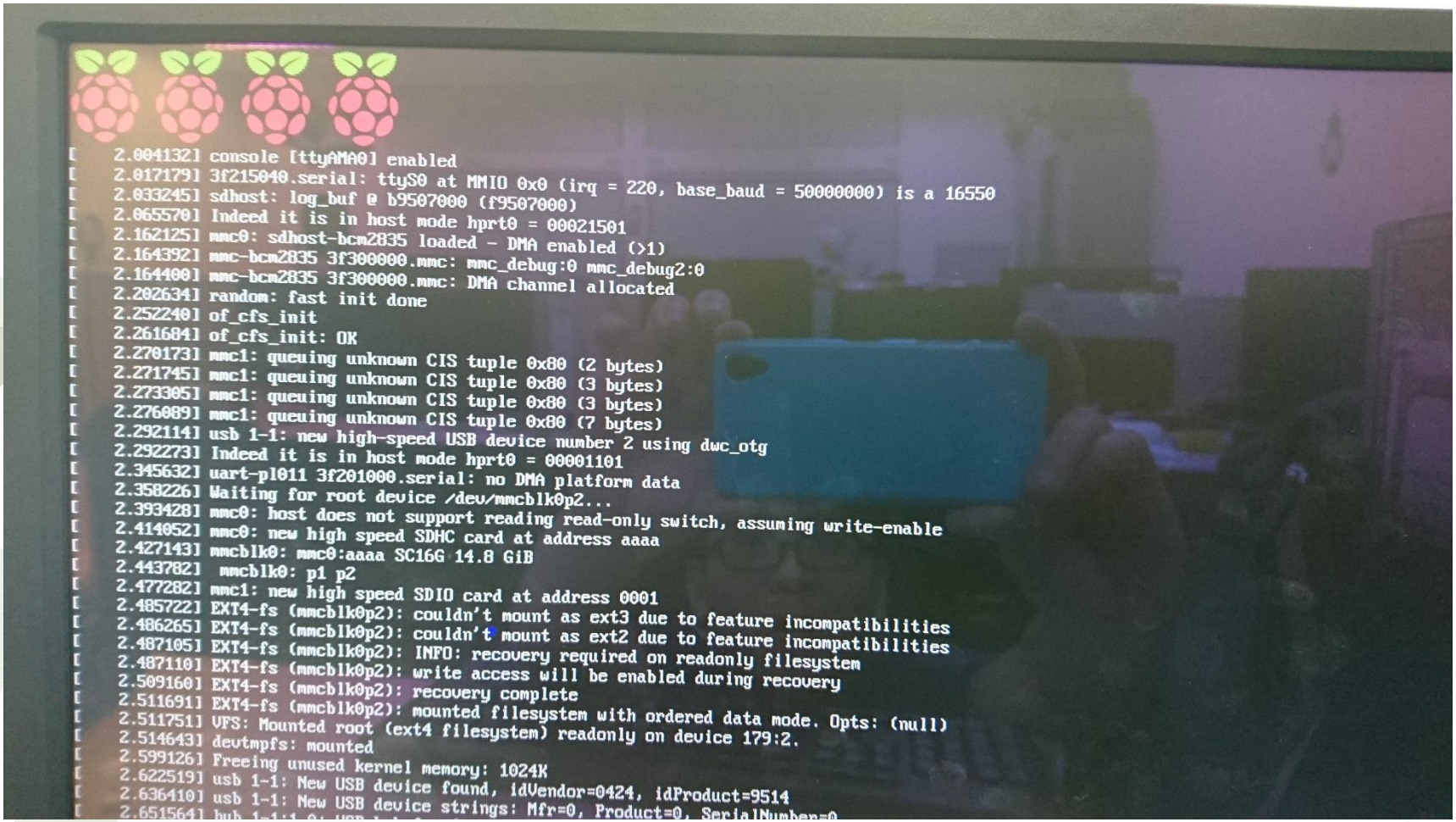
- root filesystem於FILESYSTEM分割區下解壓縮



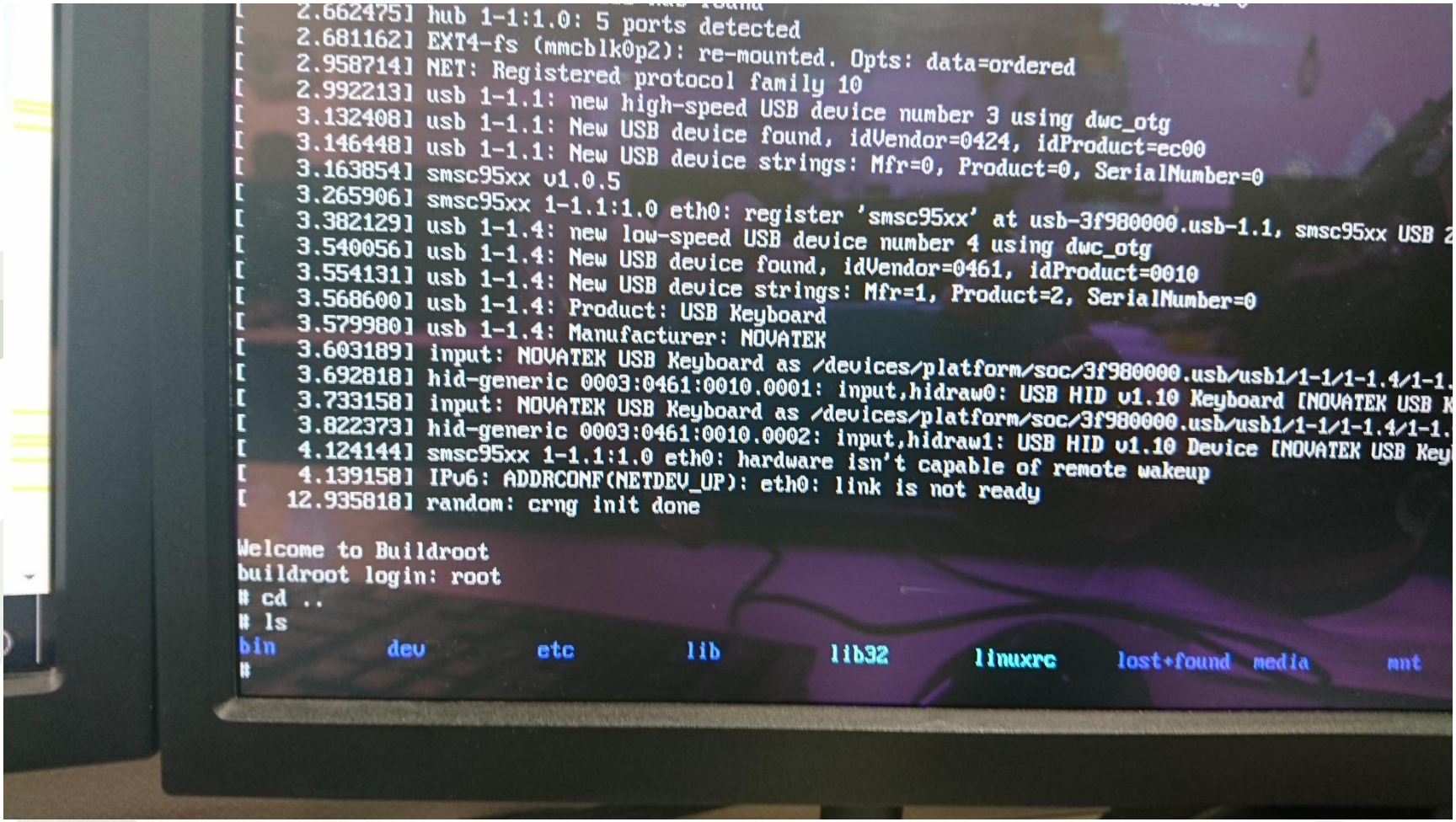


Reference from https://learn.littlebirdelectronics.com.au/guides/getting-started-with-the-raspberry-pi-3b

開機與測試 (1/2)


[2.004132] console [ttyAMA0] enabled
[2.017179] 3f215040.serial: ttyS0 at MMIO 0x0 (irq = 220, base_baud = 50000000) is a 16550
[2.033245] sdhost: log_buf @ b9507000 (f9507000)
[2.065570] Indeed it is in host mode hprt0 = 00021501
[2.162125] mmc0: sdhost-bcm2835 loaded - DMA enabled (>1)
[2.164392] mmc-bcm2835 3f300000.mmc: mmc_debug:0 mmc_debug2:0
[2.164400] mmc-bcm2835 3f300000.mmc: DMA channel allocated
[2.202634] random: fast init done
[2.252240] of_cfs_init
[2.261684] of_cfs_init: OK
[2.270173] mmc1: queuing unknown CIS tuple 0x80 (2 bytes)
[2.271745] mmc1: queuing unknown CIS tuple 0x80 (3 bytes)
[2.273305] mmc1: queuing unknown CIS tuple 0x80 (3 bytes)
[2.276089] mmc1: queuing unknown CIS tuple 0x80 (7 bytes)
[2.292114] usb 1-1: new high-speed USB device number 2 using dwc_otg
[2.292273] Indeed it is in host mode hprt0 = 00001101
[2.345632] uart-pl011 3f201000.serial: no DMA platform data
[2.358226] Waiting for root device /dev/mmcblk0p2...
[2.393428] mmc0: host does not support reading read-only switch, assuming write-enable
[2.414052] mmc0: new high speed SDHC card at address aaaa
[2.427143] mmcblk0: mmc0:aaaa SC16G 14.8 GiB
[2.443782] mmcblk0: p1 p2
[2.477282] mmc1: new high speed SDIO card at address 0001
[2.485722] EXT4-fs (mmcblk0p2): couldn't mount as ext3 due to feature incompatibilities
[2.486265] EXT4-fs (mmcblk0p2): couldn't mount as ext2 due to feature incompatibilities
[2.487105] EXT4-fs (mmcblk0p2): INFO: recovery required on readonly filesystem
[2.487110] EXT4-fs (mmcblk0p2): write access will be enabled during recovery
[2.509160] EXT4-fs (mmcblk0p2): recovery complete
[2.511691] EXT4-fs (mmcblk0p2): mounted filesystem with ordered data mode. Opts: (null)
[2.511751] VFS: Mounted root (ext4 filesystem) readonly on device 179:2.
[2.514643] devtmpfs: mounted
[2.599126] Freeing unused kernel memory: 1024K
[2.622519] usb 1-1: New USB device found, idVendor=0424, idProduct=9514
[2.636410] usb 1-1: New USB device strings: Mfr=0, Product=0, SerialNumber=0

開機與測試 (2/2)


[2.662475] hub 1-1:1.0: 5 ports detected
[2.681162] EXT4-fs (mmcblk0p2): re-mounted. Opts: data=ordered
[2.958714] NET: Registered protocol family 10
[2.992213] usb 1-1.1: new high-speed USB device number 3 using dwc_otg
[3.132408] usb 1-1.1: New USB device found, idVendor=0424, idProduct=ec00
[3.146448] usb 1-1.1: New USB device strings: Mfr=0, Product=0, SerialNumber=0
[3.163854] smsc95xx v1.0.5
[3.265906] smsc95xx 1-1.1:1.0 eth0: register 'smsc95xx' at usb-3f980000.usb-1.1, smsc95xx USB
[3.382129] usb 1-1.4: new low-speed USB device number 4 using dwc_otg
[3.540056] usb 1-1.4: New USB device found, idVendor=0461, idProduct=0010
[3.554131] usb 1-1.4: New USB device strings: Mfr=1, Product=2, SerialNumber=0
[3.568600] usb 1-1.4: Product: USB Keyboard
[3.579980] usb 1-1.4: Manufacturer: NOVATEK
[3.603189] input: NOVATEK USB Keyboard as /devices/platform/soc/3f980000.usb/usb1/1-1/1-1.4/1-1.
[3.692818] hid-generic 0003:0461:0010.0001: input,hidraw0: USB HID v1.10 Keyboard [NOVATEK USB K
[3.733158] input: NOVATEK USB Keyboard as /devices/platform/soc/3f980000.usb/usb1/1-1/1-1.4/1-1.
[3.822373] hid-generic 0003:0461:0010.0002: input,hidraw1: USB HID v1.10 Device [NOVATEK USB Key
[4.124144] smsc95xx 1-1.1:1.0 eth0: hardware isn't capable of remote wakeup
[4.139158] IPv6: ADDRCONF(NETDEV_UP): eth0: link is not ready
[12.935818] random: crng init done
Welcome to Buildroot
buildroot login: root
cd ..
ls
bin dev etc lib lib32 linuxrc lost+found media mnt
#

Reference

- http://fichugh.blogspot.tw/2016/02/buildroot-study.html
- http://www.embeddedforu.com/embedded-linux/raspberry-pi/embedded-linux-development-on-raspberry-pi-using-buildroot-part1/
- http://www.embeddedforu.com/embedded-linux/raspberry-pi/embedded-linux-development-on-raspberry-pi-using-buildroot-part2-2/

DEMO Target

- 確實使用Buildroot建構出最簡易Linux系統，並成功開機且使用root權限進入根目錄
- BusyBox部分，需要 (包含) 選擇編譯下列工具程式
  - gzip
  - bzip2
  - top (or atop)
  - vim (editor)

（註：請確實確定能完全成功地開機，而不是進入initramfs）

Q & A (請於實驗報告裡回覆)

- 請問下面firmware目錄下的各個檔案的用途分別是什麼?
  - output/rpi-firmware/
    - bootcode.bin
    - cmdline.txt
    - cofing.txt
    - fixup.dat
    - start.elf
- "cmdline.txt"的內容是什麼，相關參數的意義是什麼?